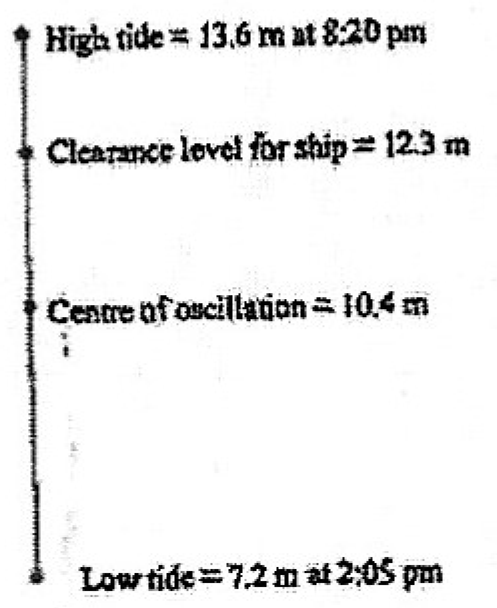
Mechanics WS 3

1. Two particles P and Q are projected from the same point O with the same velocity 25 m/s. They both strike the horizontal plane through O at the point A, 60 m from O. P reaches A before Q. Show that:
   a. the angle of projection of P is $\tan^{-1}\frac{4}{3}$ and that of Q is $\tan^{-1}\frac{4}{3}$.
   b. the time of flight of P is 3 seconds.
   c. the distance between P and Q at the instant P reaches A is $15\sqrt{2}$m.
   d. the ratio of the maximum heights reached by P and Q is $\frac{9}{16}$.
2. A particle of unit mass falls vertically from reset in a medium with resistive force $R = kv$ where v is the velocity of the particle at time t. (k is a constant). Find the velocity v and hence show that the terminal velocity is given by $\frac{g}{k}$.
3. An object is projected vertically upwards with initial velocity U from the earch's surface. The acceleration obeys the law given by $\frac{d^2y}{dt^2} = -\frac{k}{x^2}$ where x is the distance of the particle from the centre of the earth whose radius is R. Given the acceleration is g when $x = R$, show that velocity v in any position is given by: $v^2 = U^2 - 2gR^2(\frac{1}{R} - \frac{1}{x})$. Hence show that $U = 12\ km/s$ for the escape velocity. (i.e. the object does not return to the earth). ($R = 6400\ km, g = 10\ m.s^{-2}$) (Hint: $x \to \infty$ for escapte from the earth and $v^2 > o$)
4. A particle moves under gravity in a resistive medium with the resistance $R = kv$, where v is the velocity in any position and k is a constant. The particle is projected vertically upwards with velocity $U = \frac{g}{k}$. Show that at any time the expressions for the velocity v and position x are given by: $v = U(2e^{-kt} - 1)$ and $x = \frac{U}{k}(2 - 2e^{-kt} - kt)$. Find the greatest height achieved by the particle. (Hint: $v = 0$ for the greatest height)
5. A particle of mass m is projected vertically upwards under gravity with velocity $U\ m/s$. The resistive force is $R9v) = mkv^2$ at any time t and position x. Show that the expression for x is given by: $x = \frac{1}{2k}\log_e[\frac{g+kU^2}{g+kv^2}]$ Show that the greatest height H is given by: $H = \frac{1}{2k}\log_e[1 + \frac{kU^2}{g}]$
6. A body of mass m falls from rest in a medium with resistive force $R = kv$, where k is the coefficient of air resistance and v is the speed of the object. (k is a constant.) Prove that the distance x fallen when the velocity is v, is given by: $x = -\frac{mv}{k} - \frac{m^2g}{k^2}\log_e[1 - \frac{kv}{mg}]$ Find the terminal velocity for a falling 70 kg sky-driver, if $k = 14$ and $g = 10\ m/s^2$. Express your answer in km/h.
7. A ball of mass m is thrown vertically upwards with velocity U. The air resistance if per unit mass $f(v) = kv^2$, at a distance x when velocity is v.
   a. Draw a neat sketch of the motion, showing the forces acting at a distance x from the pint of projection.
   b. Show that $x = \frac{1}{2k}\log_e[\frac{g+kU^2}{g+kv^2}]$
   c. Show that the greatest height achieved is $H = \frac{1}{2k}\log_e[1 + \frac{k}{g} \cdot U^2]$
   d. Draw a neat sketch of the downward motion of this ball after it reaches the greatest height H. Show that the distance y fallen when velocity is W, is given by $y = \frac{1}{2k}\log_e[\frac{g}{g-kW^2}]$
   e. Deduce from (d) that the terminal velocity V is given by $V^2 = \frac{g}{k}$, and that H can also be given by: $H = \frac{1}{2k}\log_e[\frac{g}{g-kW^2}]$ where W is the velocity when $y = H$ for downward motion.
   f. From (c) and (d) deduce that $\frac{1}{U^2} + \frac{1}{V^2} = \frac{1}{W^2}$

8. A particle is thrown vertically upwards where the air resistance is given by $R = 0.01mv^2$. If the velocity of projection is $60\ m/s$, find the
   a. time to reach the highest point
   b. greatest height H.
9. A particle falling from rest in a vertical line in a medium with resistance kv^2 per unit mass, k is a constant, v is the velocity at any time and position x, prove that it acquires a speed $[\sqrt{1-e^{-2kh}}]v_T$. In falling through a distance h, where v_T is the terminal velocity given by $\sqrt{\frac{g}{k}}$. A particle projected upwards I the same medium with initial speed $v_I$and returns to the point of projection with speed V_R. Prove that $v_R^{-2} = v_I^{-2} + v_T^{-2}$
10. The depth of the water in a harbour is 7.2 m at low water and 13.6 m at high water. On Monday, low water is at 2:05 pm and high water is at 8:20 pm. A particular ship requires water to a depth of at least 12.3 m to leave harbour. The ship's captain wants to leave harbour between noon and midnight. Find between what times the ship can leave, assuming that the motion of the tide is simple harmonic.



11. A particle of unit mass falls from rest in a medium with resistive force $R = kv^2$, where k is a constant. Prove that the distance x fallen when the velocity is v, is given by: $x = \frac{1}{2k}\log_e(\frac{g}{g-kv^2})$. Find the distance fallen when it reaches half of its terminal velocity, (Hint: The terminal velocity is given by $g - kv^2 = 0$, i.e. $v = \sqrt{g/k}$).
12. A particle moves in a straight line so that when it is x metres from the origin O its velocity v m/s is given by $\mathrm{v}^2 = 32 + 8\mathrm{x} - 4\mathrm{x}^2$
    a. Prove that the particle is moving in SHM
    b. Find the centre of motion
    c. Find the period and amplitude
    d. Given that the particle is initially at $\mathrm{x} = 4$, which of the 2 equations could describe its motion: $\mathrm{x} = 1 + 3\sin 2\mathrm{t}$ or $\mathrm{x} = 1 + 3\cos 2\mathrm{t}$
    e. When does the particle pass through the origin for the first time?
13. On a certain day the depth of water in a harbour at low tide at 12 noon is 4 metres. At the following high tide the depth of water is 11 metres and the interval between successive high tides is 12 hours 20 minutes. Assuming the rise and fall of the surface of the water to be simple harmonic, find the earliest time after 12 noon that a ship may safely enter the harbour if the minimum depth of 9 metres of water is required.
14. The rise and fall of the tide at the port is in SHM. The time interval between high tides is 8 hours. The port entrance has a depth of 8m at high tide and 2m at low tide. On Monday morning low tide is at noon. What is the earliest time on Monday that a ship needing 6m of water can pass through the entrance?

Answers:

2. $v = \frac{g}{k}(1 - e^{-kt})$
4. $\frac{U}{k}(1 - \log_e 2)$
6. $180\ km/h.$
8. (a) $3.43\ s$ (b) $76.3m$
10. $6{:}28pm - 10{:}12pm$
11. $\frac{1}{2k}\log_e(4/3)$
12. $x = 1, \pi, 3, t = 0.953$
13. 3:57pm
14. 2:26 pm